RC car — ESP32-S3-CAM brain + RP2040 PWM splitter

U1 = Freenove L1–L20 / R1–R20

U2 = Pico 1–40 as on board

U3/U4 = ROB-14450 JP1-x/JP2-x

MCU BLOCKS

ESP32-S3-CAM (N16R8) — Freenove 2x20 header
U1

+3V3_S3	L1	R1	S3_TXD
	L2 3V3	R2 GPIO43_U0TXD	LINK_RX_S3
	L3 RST	R3 GPIO44_U0RXD	I2C_SDA
	L4 GPIO4_CAM_SIOD	R4 GPIO1_ADC1_CH0	I2C_SCL
	L5 GPIO5_CAM_SIOC	R5 GPIO2_ADC1_CH1	
	L6 GPIO6_CAM_VSYNC	R6 GPIO42_MTMS	
	L7 GPIO7_CAM_HREF	R7 GPIO41_MTDI	
	L8 GPIO15_CAM_XCLK	R8 GPIO40_SD_DATA	
	L9 GPIO16_CAM_Y9	R9 GPIO39_SD_CLK	
	L10 GPIO17_CAM_Y8	R10 GPIO38_SD_CMD	
	L11 GPIO18_CAM_Y7	R11 GPIO37_PSRAM*	
	L12 GPIO8_CAM_Y4	R12 GPIO36_PSRAM*	
	L13 GPIO3_JTAG_EN	R13 GPIO35_PSRAM*	
	L14 GPIO46_LOG	R14 GPIO0_BOOT	
	L15 GPIO9_CAM_Y3	R15 GPIO45_VSPI	S3_LED
	L16 GPIO10_CAM_Y5	R16 GPIO48_WS2812	I2S_BCK
	L17 GPIO11_CAM_Y2	R17 GPIO47	I2S_DIN
	L18 GPIO12_CAM_Y6	R18 GPIO21	I2S_LRCK
	L19 GPIO13_CAM_PCLK	R19 GPIO20_USB_D+	
+5V	L20 GPIO14 5V	R20 GPIO19_USB_D− GND	GND

Raspberry Pi Pico (RP2040) — 1-40 header
U2

DRV1_PWMA	1 GP0	40 VBUS	
DRV1_AIN1	2 GP1	39 VSYS	VSYS
GND	3 GND	38 GND	GND
DRV1_AIN2	4 GP2	37 3V3_EN	
DRV1_BIN2	5 GP3	36 3V3_out	+3V3_PICO
DRV1_BIN1	6 GP4	35 ADC_VREF	
DRV1_PWMB	7 GP5	34 GP28_ADC2	STBY_PICO
GND	8 GND	33 AGND	GND
DRV2_PWMA	9 GP6	32 GP27_ADC1	
DRV2_AIN1	10 GP7	31 GP26_ADC0	
DRV2_AIN2	11 GP8	30 RUN	
DRV2_BIN2	12 GP9	29 GP22	
GND	13 GND	28 GND	GND
DRV2_BIN1	14 GP10	27 GP21	
DRV2_PWMB	15 GP11	26 GP20	
	16 GP12	25 GP19	
	17 GP13	24 GP18	
GND	18 GND	23 GND	GND
	19 GP14	22 GP17	LINK_RX_PICO
	20 GP15	21 GP16	PICO_TXD

SparkFun ROB-14450 #1 (left: FL=A, RL=B)
U3

GND	JP1-1 GND	PWMA JP2-1	DRV1_PWMA
MOT_RL_A	JP1-2 B01	AIN2 JP2-2	DRV1_AIN2
MOT_RL_B	JP1-3 B02	AIN1 JP2-3	DRV1_AIN1
MOT_FL_B	JP1-4 A02	STBY JP2-4	MOT_STBY
MOT_FL_A	JP1-5 A01	BIN1 JP2-5	DRV1_BIN1
GND	JP1-6 GND	BIN2 JP2-6	DRV1_BIN2
+3V3_PICO	JP1-7 VCC	PWMB JP2-7	DRV1_PWMB
VM_MOTOR	JP1-8 VM	GND JP2-8	GND

SparkFun ROB-14450 #2 (right: RR=A, FR=B)
U4

GND	JP1-1 GND	PWMA JP2-1	DRV2_PWMA
MOT_FR_A	JP1-2 B01	AIN2 JP2-2	DRV2_AIN2
MOT_FR_B	JP1-3 B02	AIN1 JP2-3	DRV2_AIN1
MOT_RR_B	JP1-4 A02	STBY JP2-4	MOT_STBY
MOT_RR_A	JP1-5 A01	BIN1 JP2-5	DRV2_BIN1
GND	JP1-6 GND	BIN2 JP2-6	DRV2_BIN2
+3V3_PICO	JP1-7 VCC	PWMB JP2-7	DRV2_PWMB
VM_MOTOR	JP1-8 VM	GND JP2-8	GND

MOTOR DRIVE + IMU

BMI160 breakout
U5

+3V3_S3	1 VDD	SDA 5	I2C_SDA
+3V3_S3	2 VDDIO	SCL 6	I2C_SCL
GND	3 GND	SDO 7	GND
+3V3_S3	4 CS	INT1 8	

GLUE

1k UART series
R1

S3_TXD	1	2 LINK_RX_PICO

1k UART series
R2

PICO_TXD	1	2 LINK_RX_S3

4.7k I2C pull-up (DNP?)
R3

+3V3_S3	1	2 I2C_SDA

4.7k I2C pull-up (DNP?)
R4

+3V3_S3	1	2 I2C_SCL

1k STBY series
R5

STBY_PICO	1	2 MOT_STBY

330R strip series
R6

S3_LED	1	2 LED_STRIP_DIN

E-STOP / VSYS

E-stop (NC, optional)
SW1

MOT_STBY	1	2 GND

BAT54 Schottky
D1

+5V	1	2 VSYS

AUDIO + LED

PCM5102A DAC (on-board 3V3 reg, VIN = 5 V)
U6

	SCK_(GND) 1	FLT_(GND) 7	
I2S_BCK	BCK 2	DEMP_(GND) 8	
I2S_DIN	DIN 3	XSMT_(3V3) 9	
I2S_LRCK	LCK 4	FMT_(GND) 10	
GND	GND 5	A3V3_(int_3V3) 11	
+5V	VIN 6	AGND 12	GND
		ROUT 13	
		AGND 14	GND
		LOUT 15	AUDIO_L

PAM8302A class-D amp
U7

+5V	1 VDD	IN+ 4	AUDIO_L
GND	2 GND	IN− 5	GND
	3 SD	OUT+ 6	SPK_A
		OUT− 7	SPK_B

WS2812 strip (5 V)
LED1

+5V	1 VDD	DIN 3	LED_STRIP_DIN
GND	2 GND		

speaker 8R
SP1

SPK_A	1	2 SPK_B

POWER

3S LiPo 11.1 V (9.0–12.6 V)
BT1

VBAT_RAW	1	2 GND

fuse 7.5 A
F1

VBAT_RAW	1	2 VBAT

XL4015 buck 5 A -> MOTOR RAIL = 8.0 V
PS1

VBAT	1 IN+	OUT+ 3	VM_MOTOR
GND	2 IN−	OUT− 4	GND

LM2596 buck 3 A -> 5.0 V LOGIC
PS2

VBAT	1 IN+	OUT+ 3	+5V
GND	2 IN−	OUT− 4	GND

MECHANICAL

motor FL
M1

MOT_FL_A	1	2 MOT_FL_B

motor RL
M2

MOT_RL_A	1	2 MOT_RL_B

motor RR
M3

MOT_RR_A	1	2 MOT_RR_B

motor FR
M4

MOT_FR_A	1	2 MOT_FR_B

Architecture and rationale: docs/circuit-plan.md
Auto-generated by hardware/gen_schematic.py — do not hand-edit

Sheet: /
File: car.kicad_sch

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